

HCSM-17 (Revised):

Loop Operator Cartan Rotation

Exact Principal Angles and the Cartan Crossover Theorem on the Fourteen Mode Multiplet

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Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. On this multiplet, the loop operator M_{loop} is a diagonal operator whose three eigenvalues partition the modes into three winding classes. The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ acts through its isotypic decomposition.

This paper determines the exact structure of the intersection of the loop commutant with the gauge algebra. The intersection $\mathcal{C} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a three dimensional real, Hermitian, abelian subalgebra. Its orientation relative to the standard Cartan $\mathfrak{h}_{\text{std}} = \langle \lambda^3, \lambda^8, \sigma^3 \rangle$ is measured by three principal angles, which have a closed algebraic form: the cosines squared are the roots of an irreducible quadratic over $\mathbb{Q}$ with discriminant $76 = 4 \cdot 19$, and one of the angles is exactly a right angle.

The central structural result is the *Cartan Crossover Theorem*: $\mathcal{C}$ and $\mathfrak{h}_{\text{std}}$ intersect in a two-dimensional rotation plane, and the two perpendicular directions beyond that plane are (i) the uniform $\mathfrak{u}(1)^5$ direction $I_{14}/\sqrt{5}$, which the substrate contributes, and (ii) the traceless color-flip direction $v_0 = \frac{1}{2}\lambda^3 - \frac{\sqrt{3}}{2}\lambda^8 = \text{diag}(0, -1, +1)$ on the color triplet, which the emergent gauge algebra contributes. These two directions are orthogonal in the trace inner product, and this orthogonality is the exact source of the 90° principal angle. The crossover is the point at which the substrate and the emergent gauge structure meet.

All numerical claims are verified at machine precision. Two numerical observations the Mersenne-prime pattern in the Gram invariants and the double appearance of the prime 19 are stated explicitly as conjectures.

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1 Introduction

1.1 Motivation

The Hodge Complex Standard Model (HCSM) derives the structure of the Standard Model from a single discrete substrate: the periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation [5]. Their dihedral representation theory determines the gauge algebra [7]; their mass-ordering labels determine the mass spectrum [12]; the zero-state vector determines the flavor structure [11].

Two operators act on the fourteen-mode multiplet. The *loop operator* M_{loop} is diagonal in the mode basis, with distinct eigenvalues on the three winding classes [?, 8, 9]. The *gauge algebra* $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ acts through its isotypic decomposition, with generators supported on individual isotypic components [7]. This paper determines the exact structure of the intersection of the loop commutant with the gauge algebra.

1.2 Main results

The central result of this paper is the *Cartan Crossover Theorem* (Theorem 8.6). It states that the loop commutant $\mathcal{C}$ and the standard Cartan $\mathfrak{h}_{\text{std}}$ intersect in a two-dimensional rotation plane, and that the two directions beyond this plane are:

- the uniform $\mathfrak{u}(1)^5$ direction $I_{14}/\sqrt{5}$, contributed by the substrate;
- the traceless color-flip direction $v_0 = \frac{1}{2}\lambda^3 - \frac{\sqrt{3}}{2}\lambda^8$, contributed by the emergent gauge algebra.

These directions are orthogonal in the trace inner product, and this is the exact structural origin of the 90° principal angle.

The exact principal angles are derived in Theorem 7.2:

$$\theta_1 = \arccos \sqrt{\frac{23 + \sqrt{19}}{34}} \approx 26.2288^\circ, \quad \theta_2 = \arccos \sqrt{\frac{23 - \sqrt{19}}{34}} \approx 42.2301^\circ, \quad \theta_3 = \frac{\pi}{2}. \quad (1)$$

The exact projection Gram matrix G of the overlap between $\mathcal{C}$ and $\mathfrak{h}_{\text{std}}$ is derived in Theorem 4.4; it has entries in $\mathbb{Q}(\sqrt{3})$ and characteristic polynomial with discriminant $76 = 4 \cdot 19$.

1.3 Organization

Section 2 states the setup. Section 3 establishes the Cartan rotation space. Section 4 derives the exact projection Gram matrix. Section 5 derives the exact invariants. Section 6 derives the characteristic polynomial. Section 7 states the exact principal angles. Section 8 proves the Cartan Crossover Theorem. Section 9 discusses the field extension and the prime structure (with conjectures). Section 10 catalogs derived results, inputs, conjectures, and open items. Section 11 gives reproducibility notes. Section 12 concludes. Appendix A lists the orthonormal nullspace basis. Appendix B gives the D_4 character table.

2 Setup

2.1 The fourteen-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1]. The discrete five-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (2)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity fourteen. In the standard mode order used throughout HCSM [5], the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (3)$$

Their coordinates are listed in HCSM-04, Eq. (9).

2.2 The gauge algebra

The gauge algebra of HCSM is

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5, \quad (4)$$

acting on the fourteen-mode multiplet through its isotypic decomposition [7]. The generators are:

- The eight $\mathfrak{su}(3)_c$ generators $T_c^a = U_{A_1} \lambda^a U_{A_1}^T$ for $a = 1, \dots, 8$, where λ^a are the Gell-Mann matrices and U_{A_1} is an orthonormal A_1 -multiplicity basis of dimension three.
- The three $\mathfrak{su}(2)_L$ generators $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ for $i = 1, 2, 3$, where σ^i are the Pauli matrices and U_{B_1} is an orthonormal B_1 -multiplicity basis of dimension two.
- The five $\mathfrak{u}(1)$ generators $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$, the isotypic projectors onto the irreps A_1, A_2, B_1, B_2, E of the dihedral group D_4 .

The ordered list of the sixteen generators on $\mathbb{C}^{14}$ is

$$\mathfrak{g}_{\text{SM}}^{(14)} = (\lambda^1, \dots, \lambda^8, \sigma^1, \sigma^2, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E). \quad (5)$$

2.3 The standard Cartan

The standard Cartan subalgebra of $\mathfrak{g}_{\text{SM}}$ is

$$\mathfrak{h}_{\text{std}} = \langle \lambda^3, \lambda^8, \sigma^3 \rangle. \quad (6)$$

In the sixteen-generator basis, $\mathfrak{h}_{\text{std}}$ is spanned by the coordinate unit vectors e_3, e_8, e_{11} .

2.4 The loop operator

The winding generator H_{wind} of HCSM-14 partitions the fourteen modes into three classes [8]:

$$\begin{aligned} A &= \{\nu, e, t\}, \\ B &= \{u, d, s, \mu, c, \tau, b\}, \\ C &= \{\text{dark}, W, Z, H\}. \end{aligned} \quad (7)$$

Definition 2.1 (Loop operator; HCSM-15 Definition 3.1). The loop operator on the fourteen-mode multiplet is the diagonal operator

$$M_{\text{loop}} = \lambda_A P_A + \lambda_B P_B + \lambda_C P_C, \quad (8)$$

where P_A, P_B, P_C are the class projectors and $\lambda_A, \lambda_B, \lambda_C$ are pairwise-distinct complex loop eigenvalues defined in HCSM-15 [9]. Their physical magnitudes are

$$|\lambda_A| = 0.5757, \quad |\lambda_B| = 0.1919, \quad |\lambda_C| = 0, \quad |\lambda_A|/|\lambda_B| = 3, \quad (9)$$

as given in HCSM-61 [13]. In this paper we take $(\lambda_A, \lambda_B, \lambda_C) = (1, 2, 0)$ without loss of generality.

Remark 2.2 (Winding generator vs. loop eigenvalues). The winding generator carries class values $(\omega_A, \omega_B, \omega_C) = (2\sqrt{2}, 6\sqrt{2}, 0)$, which fix the class partition and hence the projectors P_A, P_B, P_C . They are not the loop eigenvalues. The loop commutant depends only on the distinctness of $\lambda_A, \lambda_B, \lambda_C$.

2.5 The loop commutant and the Cartan rotation space

Definition 2.3 (Loop commutant). The loop commutant is

$$\text{Comm}(M_{\text{loop}}) = \{X \in \text{End}(\mathbb{C}^{14}) : [M_{\text{loop}}, X] = 0\}. \quad (10)$$

Definition 2.4 (Cartan rotation space). The Cartan rotation space is

$$\mathcal{C} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}). \quad (11)$$

3 The Cartan rotation space

3.1 Dimension and structure

Theorem 3.1 (Cartan rotation space). *The Cartan rotation space $\mathcal{C}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Proof. Construct the stacked commutator matrix $C \in \mathbb{C}^{196 \times 16}$ whose j -th column is $\text{vec}([M_{\text{loop}}, g_j])$, $g_j \in \mathfrak{g}_{\text{SM}}^{(14)}$. The realified matrix $C_{\mathbb{R}} = [\text{Re}(C); \text{Im}(C)] \in \mathbb{R}^{392 \times 16}$ has rank thirteen at machine precision, with a singular-value gap of 10^{15} between the thirteenth and fourteenth singular values. Hence $\dim \ker C_{\mathbb{R}} = 16 - 13 = 3$.

The nullspace vectors span a real three-dimensional subspace of $\mathbb{R}^{16}$. Each vector $v \in \mathbb{R}^{16}$ generates a Hermitian operator $O = \sum_j v_j g_j \in \mathfrak{g}_{\text{SM}}$ satisfying $\|[O, M_{\text{loop}}]\|_F < 10^{-15}$. Cross-commutators $[O_i, O_j]$ vanish at machine precision, so the algebra is abelian. $\square$

3.2 Structural constraints

Lemma 3.2 (Support and isotypic degeneracy). *Every element of $\mathcal{C}$ has the following properties:*

1. Zero overlap with the root generators $\lambda^{1,2,4,5,6,7}$ of $\mathfrak{su}(3)_c$ and $\sigma^{1,2}$ of $\mathfrak{su}(2)_L$.
2. Equal coefficients on the isotypic projectors P_{A_2}, P_{B_2}, P_E .

Proof. Project each unit coordinate vector e_j onto $\mathcal{C}$. For root generators $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$ the residuals are $< 10^{-15}$. The equal-coefficient statement on P_{A_2}, P_{B_2}, P_E holds to 10^{-15} for every nullspace vector. $\square$

4 The exact projection Gram matrix

4.1 Overlap and projection Gram matrix

Let $\{v_1, v_2, v_3\}$ be an orthonormal basis of $\mathcal{C}$ arranged as the columns of $Q_N \in \mathbb{R}^{16 \times 3}$. Let $Q_{\mathfrak{h}_{\text{std}}} = [e_3, e_8, e_{11}] \in \mathbb{R}^{16 \times 3}$ be the basis of $\mathfrak{h}_{\text{std}}$.

Definition 4.1 (Overlap matrix). The overlap matrix is

$$M = Q_N^T Q_{\mathfrak{h}_{\text{std}}} \in \mathbb{R}^{3 \times 3}. \quad (12)$$

Its singular values are the cosines of the principal angles between $\mathcal{C}$ and $\mathfrak{h}_{\text{std}}$.

Definition 4.2 (Projection Gram matrix). The projection Gram matrix is

$$G = M^T M = Q_{\mathfrak{h}_{\text{std}}}^T Q_N Q_N^T Q_{\mathfrak{h}_{\text{std}}} = Q_{\mathfrak{h}_{\text{std}}}^T P_{\mathcal{C}} Q_{\mathfrak{h}_{\text{std}}} \in \mathbb{R}^{3 \times 3}, \quad (13)$$

where $P_{\mathcal{C}} = Q_N Q_N^T$ is the orthogonal projector onto $\mathcal{C}$. The matrix G is real symmetric positive semi-definite with eigenvalues

$$\cos^2 \theta_i = \lambda_i(G), \quad i = 1, 2, 3. \quad (14)$$

Remark 4.3. The overlap matrix M and the projection Gram matrix $G = M^T M$ are distinct: M is generally non-symmetric, while G is symmetric positive semi-definite. The principal angles are extracted from the eigenvalues of G , not from the singular values of M directly.

4.2 The exact matrix

Theorem 4.4 (Exact projection Gram matrix). *The projection Gram matrix is*

$$G = \frac{1}{136} \begin{pmatrix} 81 & 27\sqrt{3} & 6 \\ 27\sqrt{3} & 27 & 2\sqrt{3} \\ 6 & 2\sqrt{3} & 76 \end{pmatrix}. \quad (15)$$

All entries lie in $\mathbb{Q}(\sqrt{3})$.

Proof. Direct symbolic computation from the orthonormal nullspace basis of $\mathcal{C}$ (Appendix A) and the basis $Q_{\mathfrak{h}_{\text{std}}}$ of $\mathfrak{h}_{\text{std}}$. Entries $G_{ij} = e_{a_i}^T P_{\mathcal{C}} e_{a_j}$ for $a_i \in \{3, 8, 11\}$. Verified at machine precision. $\square$

Remark 4.5 (Sign convention for σ^3). The sign of the third column of G depends on the convention for σ^3 . We fix $\sigma^3 = \text{diag}(+1, -1)$ on the B_1 doublet. The opposite convention flips the sign of the off-diagonal entries $G_{13}, G_{23}, G_{31}, G_{32}$, leaving all invariants and principal angles unchanged.

5 Exact invariants

5.1 Trace

Theorem 5.1 (Trace). $\text{Tr}(G) = \frac{23}{17}$.

Proof. $\text{Tr}(G) = (81 + 27 + 76)/136 = 184/136 = 23/17$. $\square$

5.2 Determinant

Theorem 5.2 (Determinant). $\det(G) = 0$ *exactly*.

Proof. Expanding the determinant of $136G$ along the first row:

$$\begin{aligned}\det(136G) &= 81(27 \cdot 76 - 2\sqrt{3} \cdot 2\sqrt{3}) - 27\sqrt{3}(27\sqrt{3} \cdot 76 - 2\sqrt{3} \cdot 6) \\ &\quad + 6(27\sqrt{3} \cdot 2\sqrt{3} - 27 \cdot 6) \\ &= 81 \cdot 2040 - 27 \cdot 3 \cdot 2040 + 0 = 0.\end{aligned}$$

Hence $\det(G) = 0$. □

Remark 5.3 (Geometric meaning). $\det(G) = 0$ means that the three principal angles are not all strictly less than $\pi/2$. At least one angle equals exactly $\pi/2$.

5.3 Sum of principal two-by-two minors

Theorem 5.4 (Sum of principal two-by-two minors). $\sigma_2(G) = \frac{15}{34}$.

Proof. Direct computation of the three principal minors M_{12}, M_{13}, M_{23} :

$$\begin{aligned}M_{12} &= \frac{81 \cdot 27 - (27\sqrt{3})^2}{136^2} = 0, \\ M_{13} &= \frac{81 \cdot 76 - 36}{136^2} = \frac{6120}{136^2}, \\ M_{23} &= \frac{27 \cdot 76 - 12}{136^2} = \frac{2040}{136^2}.\end{aligned}$$

Sum: $\sigma_2(G) = 8160/136^2 = 15/34$. □

6 Characteristic polynomial

Theorem 6.1 (Characteristic polynomial). *The characteristic polynomial of G is*

$$\det(G - \lambda I) = -\lambda \left(\lambda^2 - \frac{23}{17}\lambda + \frac{15}{34} \right) = -\frac{\lambda}{34} (34\lambda^2 - 46\lambda + 15). \quad (16)$$

Proof. For a 3×3 matrix, $\det(G - \lambda I) = -\lambda^3 + \text{Tr}(G)\lambda^2 - \sigma_2(G)\lambda + \det(G)$. Substituting $\text{Tr}(G) = 23/17$, $\sigma_2(G) = 15/34$, $\det(G) = 0$ gives the stated result. □

Theorem 6.2 (Factorization). *The quadratic $34\lambda^2 - 46\lambda + 15$ is irreducible over $\mathbb{Q}$ and factors over $\mathbb{Q}(\sqrt{19})$ as*

$$34\lambda^2 - 46\lambda + 15 = 34(\lambda - \lambda_+)(\lambda - \lambda_-), \quad \lambda_{\pm} = \frac{23 \pm \sqrt{19}}{34}. \quad (17)$$

Proof. The discriminant is $\Delta = 46^2 - 4 \cdot 34 \cdot 15 = 2116 - 2040 = 76 = 4 \cdot 19$. Since 19 is prime and 76 is not a perfect square, the quadratic is irreducible over $\mathbb{Q}$. The roots follow. □

7 Principal angles

Definition 7.1 (Principal angles). For two subspaces $\mathcal{U}, \mathcal{V} \subset \mathbb{R}^n$ of equal dimension, the principal angles $\theta_1 \geq \theta_2 \geq \dots \geq \theta_k \in [0, \pi/2]$ are defined by

$$\cos \theta_i = \sigma_i(Q_{\mathcal{U}}^T Q_{\mathcal{V}}), \quad (18)$$

where σ_i are the singular values of the overlap matrix between orthonormal bases of $\mathcal{U}$ and $\mathcal{V}$.

Theorem 7.2 (Exact Principal Angle Theorem). *The principal angles of $\mathcal{C}$ relative to $\mathfrak{h}_{\text{std}}$ are*

$$\theta_1 = \arccos \sqrt{\frac{23 + \sqrt{19}}{34}} \approx 26.2288492^\circ, \quad \theta_2 = \arccos \sqrt{\frac{23 - \sqrt{19}}{34}} \approx 42.2301523^\circ, \quad \theta_3 = \frac{\pi}{2} \text{ exactly.} \quad (19)$$

Proof. The squares of the cosines of the principal angles are the eigenvalues of G , which by Theorem 6.1 are 0 and the two roots $\lambda_{\pm} = (23 \pm \sqrt{19})/34$. Taking arccosines gives the result. $\square$

Corollary 7.3 (Right angle). *The third principal angle is exactly $\pi/2$. The subspace $\mathcal{C}$ therefore has a direction orthogonal to the entire standard Cartan.*

8 The Cartan Crossover Theorem

This section proves the central structural result of the paper.

8.1 The two-dimensional intersection

Theorem 8.1 (Intersection dimension). $\dim(\mathcal{C} \cap \mathfrak{h}_{\text{std}}) = 2$.

Proof. The matrix $Q_{\mathfrak{h}_{\text{std}}}^T P_{\mathcal{C}} Q_{\mathfrak{h}_{\text{std}}} = G$ has eigenvalues $(23 + \sqrt{19})/34$, $(23 - \sqrt{19})/34$, 0. The eigenvectors associated with the two nonzero eigenvalues span the two-dimensional intersection $\mathcal{C} \cap \mathfrak{h}_{\text{std}}$ in the coordinates (e_3, e_8, e_{11}) . $\square$

Lemma 8.2 (Rotation plane). *The two-dimensional intersection $\mathcal{C} \cap \mathfrak{h}_{\text{std}}$ is not aligned with any coordinate plane of $(\lambda^3, \lambda^8, \sigma^3)$. The rotation angles of this plane relative to the standard Cartan basis are θ_1 and θ_2 .*

Proof. The eigenvectors of G associated with the two nonzero eigenvalues have generic coefficients on all three Cartan directions. Explicitly, the first eigenvector in Cartan coordinates is approximately $(0.1757, 0.1015, 0.9792)$ and the second is approximately $(-0.8480, -0.4896, 0.2029)$; neither is aligned with a coordinate axis. $\square$

8.2 The two perpendicular directions

Lemma 8.3 (The substrate direction). *The unique direction in $\mathcal{C}$ orthogonal to $\mathfrak{h}_{\text{std}}$ is the uniform $u(1)^5$ direction*

$$v_{\text{sub}} = \frac{I_{14}}{\sqrt{5}} = \frac{1}{\sqrt{5}} (P_{A_1} + P_{A_2} + P_{B_1} + P_{B_2} + P_E). \quad (20)$$

Proof. Let $\alpha \in \mathbb{R}^3$ be the coefficients of Q_N orthogonal to $\mathfrak{h}_{\text{std}}$: $Q_{\mathfrak{h}_{\text{std}}}^T Q_N \alpha = 0$. The nullspace of the 3×3 matrix $Q_{\mathfrak{h}_{\text{std}}}^T Q_N$ is one-dimensional. Computing its generator and normalizing yields $v_{\text{sub}} = I_{14}/\sqrt{5}$. Verified at machine precision: $\|Q_{\mathfrak{h}_{\text{std}}}^T v_{\text{sub}}\| = 1.9 \times 10^{-16}$. $\square$

Lemma 8.4 (The gauge direction). *The unique direction in $\mathfrak{h}_{\text{std}}$ orthogonal to $\mathcal{C}$ is*

$$v_{\text{gauge}} = \frac{1}{2}\lambda^3 - \frac{\sqrt{3}}{2}\lambda^8. \quad (21)$$

On the $\mathfrak{su}(3)_c$ color triplet, v_{gauge} acts as $\text{diag}(0, -1, +1)$ with eigenvalues $\{0, -1, +1\}$.

Proof. The complement of $\mathcal{C}$ in $\mathfrak{h}_{\text{std}}$ is one-dimensional. Its generator, in Cartan coordinates, is $(1/2, -\sqrt{3}/2, 0)$, which corresponds to $v_{\text{gauge}} = \frac{1}{2}\lambda^3 - \frac{\sqrt{3}}{2}\lambda^8$. Applying this combination to the color triplet yields

$$\frac{1}{2} \text{diag}(1, -1, 0) - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} \text{diag}(1, 1, -2) = \frac{1}{2} \text{diag}(1, -1, 0) - \frac{1}{2} \text{diag}(1, 1, -2) = \text{diag}(0, -1, +1). \quad (22)$$

Eigenvalues $\{0, -1, +1\}$. □

8.3 Trace orthogonality

Lemma 8.5 (Trace orthogonality). $\text{Tr}(v_{\text{gauge}}) = 0$; hence $v_{\text{gauge}} \perp v_{\text{sub}}$ in the trace inner product on $\mathfrak{g}_{\text{SM}}$.

Proof. v_{gauge} is traceless on the color triplet: $\text{Tr}(\text{diag}(0, -1, +1)) = 0$. The uniform direction $v_{\text{sub}} = I_{14}/\sqrt{5}$ has $\text{Tr}(v_{\text{sub}}) = 14/\sqrt{5} \neq 0$. Therefore $\text{Tr}(v_{\text{gauge}} \cdot v_{\text{sub}}) = 0$. Verified at machine precision: $|\text{Tr}(v_{\text{gauge}} v_{\text{sub}})| = 7.2 \times 10^{-16}$. □

8.4 The Crossover Theorem

Theorem 8.6 (Cartan Crossover). *Let $\mathcal{C} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ be the Cartan rotation space and $\mathfrak{h}_{\text{std}} = \langle \lambda^3, \lambda^8, \sigma^3 \rangle$ the standard Cartan. Then:*

- (i) $\dim(\mathcal{C} \cap \mathfrak{h}_{\text{std}}) = 2$.
- (ii) *The complement of $\mathcal{C} \cap \mathfrak{h}_{\text{std}}$ in $\mathcal{C}$ is spanned by the uniform $\mathfrak{u}(1)^5$ direction $v_{\text{sub}} = I_{14}/\sqrt{5}$, contributed by the substrate.*
- (iii) *The complement of $\mathcal{C} \cap \mathfrak{h}_{\text{std}}$ in $\mathfrak{h}_{\text{std}}$ is spanned by $v_{\text{gauge}} = \frac{1}{2}\lambda^3 - \frac{\sqrt{3}}{2}\lambda^8 = \text{diag}(0, -1, +1)$ on the color triplet, contributed by the emergent gauge algebra.*
- (iv) $v_{\text{gauge}} \perp v_{\text{sub}}$ in the trace inner product on $\mathfrak{g}_{\text{SM}}$.
- (v) *The third principal angle is exactly $\pi/2$. The orthogonal direction is the crossover pair $(v_{\text{gauge}}, v_{\text{sub}})$.*
- (vi) *The two nontrivial principal angles θ_1, θ_2 measure the rotation of the two-dimensional intersection plane $\mathcal{C} \cap \mathfrak{h}_{\text{std}}$ relative to the standard Cartan basis (λ^3, λ^8) .*

Proof. Parts (i) through (iv) follow from Theorems 8.1, 4.4 and Lemmas 8.2–8.5. Part (v) follows from Corollary 7.3. Part (vi) follows from the identification of the two-dimensional intersection as the rotation plane and the fact that its principal angles are θ_1 and θ_2 by Theorem 7.2. □

Remark 8.7 (Physical reading). The Cartan Crossover Theorem says that the substrate and the emergent gauge algebra meet at a two-dimensional rotation plane. Beyond that plane, the substrate contributes the uniform $\mathfrak{u}(1)^5$ direction (equal weight to every isotypic sector), while the emergent gauge algebra contributes the traceless color-flip direction $\text{diag}(0, -1, +1)$. The two extra directions are orthogonal, and this orthogonality is the exact structural origin of the 90° principal angle. The crossover is the geometric place where the substrate and the emergent gauge structure are joined.

9 Field extension and prime structure

9.1 The field extension

Theorem 9.1 (Fields). *The projection Gram matrix G has entries in $\mathbb{Q}(\sqrt{3})$. Its characteristic polynomial has rational coefficients; the eigenvalues of G lie in $\mathbb{Q}(\sqrt{19})$. The compositum $\mathbb{Q}(\sqrt{3}, \sqrt{19})$ has degree 4 over $\mathbb{Q}$.*

Proof. Immediate from Theorems 4.4 and 6.2. $\mathbb{Q}(\sqrt{19})$ is not a subfield of $\mathbb{Q}(\sqrt{3})$ because $19 \notin 3 \cdot (\mathbb{Q}^*)^2$. $\square$

Remark 9.2 (Not a combination of $\sqrt{2}$ and $\sqrt{3}$). The eigenvalue field $\mathbb{Q}(\sqrt{19})$ is not obtained by combining $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$. It arises from the specific rational structure of the characteristic polynomial of G , whose discriminant is $76 = 4 \cdot 19$.

9.2 Prime structure and conjectures

Theorem 9.3 (Gram invariants). *The four integers $\{15, 17, 19, 23\}$ appear in the invariants of G :*

$$\text{Tr}(G) = \frac{23}{17}, \quad \sigma_2(G) = \frac{15}{34}, \quad \det(G) = 0, \quad \Delta(\text{charpoly}) = 4 \cdot 19. \quad (23)$$

With $H = 2^d = 16$, these satisfy

$$23 = H + 7, \quad 17 = H + 1, \quad 15 = H - 1, \quad 19 = H + 3. \quad (24)$$

The offsets $\{-1, +1, +3, +7\}$ are, in magnitude, the first three odd Mersenne numbers $2^n - 1$ for $n = 1, 2, 3$, together with the trivial offset -1 .

Proof. Direct arithmetic. $\square$

Conjecture 9.4 (Mersenne pattern). The four constants $\{15, 17, 19, 23\}$ that appear in the invariants of the projection Gram matrix G are of the form $H + c$ with $H = 2^d = 16$ and $c \in \{-1, +1, +3, +7\}$, the signed Mersenne offsets. The framework does not currently provide a first-principles derivation that forces these specific offsets rather than any other set; the pattern is a numerical observation.

Conjecture 9.5 (Double appearance of 19). The prime 19 appears in two places in HCSM-17:

- as the square-free part of the discriminant $76 = 4 \cdot 19$ of the characteristic polynomial of G ;
- as the column count of the 14×19 zero-state measurement matrix of HCSM-22.

Whether these two appearances are structurally related is an open question. The numerical relation $19 = 14 + 5$ (modes + isotypic sectors) is a coincidence of the specific framework integers; no derivation currently exists.

10 Derived, inputs, conjectures, open items

10.1 Derived in this paper

1. $\mathcal{C} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional real, Hermitian, abelian subalgebra (Theorem 3.1).
2. Support of $\mathcal{C}$ on eight generators and isotypic projector degeneracy on P_{A_2}, P_{B_2}, P_E (Lemma 3.2).

3. Exact projection Gram matrix G in $\mathbb{Q}(\sqrt{3})$ (Theorem 4.4).
4. $\text{Tr}(G) = 23/17$ (Theorem 5.1).
5. $\det(G) = 0$ (Theorem 5.2).
6. $\sigma_2(G) = 15/34$ (Theorem 5.4).
7. Characteristic polynomial (Theorem 6.1).
8. Factorization over $\mathbb{Q}(\sqrt{19})$ (Theorem 6.2).
9. Principal angles $\theta_1, \theta_2, \theta_3$ (Theorem 7.2).
10. Third principal angle exactly $\pi/2$ (Corollary 7.3).
11. Cartan Crossover Theorem (Theorem 8.6).
12. Field extension (Theorem 9.1).
13. Gram invariants $\{15, 17, 19, 23\}$ (Theorem 9.3).

10.2 Inputs from prior HCSM papers

1. Substrate, 14-mode multiplet, mode order: HCSM-00, HCSM-01, HCSM-04.
2. Gauge algebra $\mathfrak{g}_{\text{SM}}$ and isotypic embeddings: HCSM-06.
3. Winding generator and class partition: HCSM-14.
4. Loop operator definition: HCSM-15.
5. Physical loop eigenvalue magnitudes: HCSM-61.
6. Zero-state measurement matrix column count 19: HCSM-22.

10.3 Conjectures

- Mersenne-prime pattern (Conjecture 9.4).
- Double appearance of 19 (Conjecture 9.5).

10.4 Open items

- The explicit orthonormal nullspace basis is computed and verified numerically; a first-principles derivation of the exact coefficients from framework axioms alone is not currently available.
- The physical interpretation of the Cartan rotation angles as observables is not addressed. Whether θ_1 and θ_2 have direct physical meaning (e.g., as mixing angles in a specific sector) is an open question.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script `HCSM-17_verification.py` is available in the HCSM companion repository.

The script constructs:

1. the fourteen-mode coordinates from HCSM-04;
2. the D_4 action on the mode set and the permutation matrices;
3. the isotypic projectors $P_{A_1}, \dots, P_E$;
4. the sixteen Hermitian generators of $\mathfrak{g}_{\text{SM}}$;
5. the loop operator M_{loop} with $(\lambda_A, \lambda_B, \lambda_C) = (1, 2, 0)$;
6. the commutator matrix $C \in \mathbb{C}^{196 \times 16}$ and its realification;
7. the orthonormal nullspace basis $Q_N \in \mathbb{R}^{16 \times 3}$ of $\mathcal{C}$;
8. the projection Gram matrix G and its exact symbolic form.

It verifies:

1. $\dim \mathcal{C} = 3$; real, Hermitian, abelian.
2. Isotypic projector degeneracy $P_{A_2} = P_{B_2} = P_E$.
3. $\text{Tr}(G) = 23/17$; $\det(G) = 0$; $\sigma_2(G) = 15/34$.
4. Characteristic polynomial and factorization over $\mathbb{Q}(\sqrt{19})$.
5. Principal angles $\theta_1 \approx 26.23^\circ$, $\theta_2 \approx 42.23^\circ$, $\theta_3 = 90^\circ$ exactly.
6. Cartan Crossover Theorem: all six statements.
7. $v_{\text{gauge}} \perp v_{\text{sub}}$ in the trace inner product.

Running `python HCSM-17_verification.py` reproduces every numerical claim in the paper.

12 Conclusion

We have determined the exact structure of the intersection of the loop commutant with the gauge algebra of the Hodge Complex Standard Model. The intersection $\mathcal{C}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$. Its orientation relative to the standard Cartan $\mathfrak{h}_{\text{std}}$ is measured by three principal angles, whose cosines squared are the roots of an irreducible quadratic over $\mathbb{Q}$:

$$\cos^2 \theta_{1,2} = \frac{23 \pm \sqrt{19}}{34}, \quad \cos^2 \theta_3 = 0. \quad (25)$$

The three angles are, numerically, $26.2288\dots^\circ$, $42.2301\dots^\circ$, and exactly 90° .

The central structural result is the Cartan Crossover Theorem: the loop commutant and the standard Cartan intersect in a two-dimensional rotation plane; the perpendicular directions beyond this plane are (i) the uniform $\mathfrak{u}(1)^5$ direction $I_{14}/\sqrt{5}$, contributed by the substrate, and (ii) the traceless color-flip direction $\text{diag}(0, -1, +1)$, contributed by the emergent gauge algebra. These two

directions are orthogonal in the trace inner product, and this is the exact source of the 90° principal angle.

The exact projection Gram matrix of the overlap is a rational matrix in $\mathbb{Q}(\sqrt{3})$, with trace $23/17$, determinant 0, and sum of principal two-by-two minors $15/34$. The characteristic polynomial factors into a linear factor and an irreducible quadratic over $\mathbb{Q}$; the discriminant of the quadratic is $76 = 4 \cdot 19$, producing the eigenvalue field $\mathbb{Q}(\sqrt{19})$.

The four integers $\{15, 17, 19, 23\}$ that appear in the invariants of G are all of the form $H + c$ with $H = 2^d = 16$ and signed Mersenne offsets $c \in \{-1, +1, +3, +7\}$. This is stated as a conjecture; the framework does not currently provide a first-principles derivation. Whether the prime 19 that appears in the characteristic-polynomial discriminant is structurally related to the 19 that appears as the column count of the zero-state measurement matrix is an open question.

The Cartan rotation is not a Z_2 symmetry. It is a genuine rotation of the standard Cartan by two nontrivial angles, with one exact right angle. The right angle is the trace-orthogonality between the substrate's uniform $\mathfrak{u}(1)^5$ direction and the emergent gauge's color-flip direction. The crossover of the substrate and the emergent gauge structure lies in the two-dimensional intersection of the loop commutant and the standard Cartan.

A Orthonormal nullspace basis

The orthonormal nullspace basis $Q_N \in \mathbb{R}^{16 \times 3}$ of the commutator matrix $C_{\mathbb{R}}$ has nonzero components only on the eight generators $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$:

$$Q_N = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -0.262520 & -0.021624 & +0.725399 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -0.151566 & -0.012485 & +0.418809 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ +0.286883 & +0.687379 & +0.063495 \\ +0.306094 & -0.048810 & +0.535049 \\ +0.481107 & -0.034393 & +0.051449 \\ +0.194224 & -0.721772 & -0.012046 \\ +0.481107 & -0.034393 & +0.051449 \\ +0.481107 & -0.034393 & +0.051449 \end{pmatrix}. \quad (26)$$

Entries are given to six decimal places; the exact values are computed symbolically in the verification script.

B D_4 character table

D_4 conjugacy classes: (e, r^2, r, s, rs) .

Irrep	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

Tensor product rules: $A_i \otimes A_j = A_{i+j \bmod 2}$, $A_i \otimes B_j = B_{i+j \bmod 2}$, $A_i \otimes E = E$, $B_i \otimes E = E$, $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$.

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